

Google Play Store Analytics Dashboard

Project Requirements Document

This document defines the complete project scope, technical requirements, business objectives, analytics goals, KPI framework, dashboard requirements, technology stack, and deployment specifications for the Google Play Store Analytics & Market Intelligence Dashboard project.

1. Project Overview

The goal of this project is to develop a professional end-to-end analytics platform for Google Play Store applications using Python and Streamlit. The system will:

- Clean and preprocess raw Play Store data
- Generate business KPIs
- Perform advanced exploratory data analysis (EDA)
- Create interactive dashboards
- Provide market intelligence insights
- Support business decision-making
- Prepare data for future machine learning models

2. Business Problem Statement

The Google Play Store ecosystem contains millions of apps across categories, pricing models, ratings, installs, and engagement levels. Developers and business stakeholders face challenges understanding:

- What drives app success
- Which categories perform best
- User engagement behavior
- Revenue opportunities
- Quality vs popularity dynamics
- Market competition
- Pricing effectiveness

The dashboard should transform raw app data into actionable business intelligence.

3. Technology Stack

Frontend: • Streamlit Backend & Processing: • Python • Pandas • NumPy Visualization: • Plotly • Matplotlib • Seaborn Machine Learning: • Scikit-learn • XGBoost Statistical Analysis: • SciPy • StatsModels Deployment: • Streamlit Cloud • Render • GitHub

4. Data Cleaning Requirements

The project must include a professional-grade data cleaning pipeline. Tasks:

- Missing value handling
- Duplicate detection and removal
- Invalid value correction
- Datatype conversion
- Text normalization
- Date parsing
- Outlier analysis
- Feature engineering

Columns requiring special cleaning:

- Reviews
- Installs
- Price
- Last Updated
- Rating
- Android Version

5. Feature Engineering Requirements

The system should generate advanced analytical features. Required features: • Revenue Estimate • Install Buckets • Rating Categories • Engagement Ratio • Price Categories • Update Age • App Popularity Score • Category Market Share • User Engagement Index

6. KPI Framework

Core KPIs: • Total Apps • Total Installs • Average Rating • Total Reviews • Free vs Paid Ratio Business KPIs: • Top Performing Categories • Highest Revenue Categories • Engagement Metrics • Growth Metrics • Quality Metrics Advanced KPIs: • Market Concentration • Revenue Opportunity Score • Success Probability Index • User Satisfaction Score

7. Dashboard Requirements

The dashboard must be interactive and professionally designed. Required pages: 1. Executive Overview 2. Category Intelligence 3. User Engagement Analysis 4. Revenue & Monetization Analytics 5. App Success Analysis 6. Data Quality Dashboard Dashboard Features: • Sidebar filters • Search functionality • Interactive visualizations • Dynamic KPIs • Download cleaned data • Responsive design • Multi-page navigation

8. Visualization Requirements

Required visualizations: • Histograms • Boxplots • Heatmaps • Scatter plots • Trend charts • Bar charts • Treemaps • Bubble charts • Correlation matrices • Donut charts All charts should: • Be interactive where possible • Include tooltips • Support filtering • Follow clean UI/UX standards

9. Business Storytelling Requirements

The dashboard should tell meaningful business stories. Key stories: • Quality vs Popularity paradox • Free app market dominance • Category growth opportunities • Engagement behavior analysis • Revenue strategy insights • Market concentration analysis Each insight should include: • Observation • Business implication • Recommended action

10. Machine Learning Readiness

The cleaned dataset should support future machine learning workflows. Potential ML tasks: • Install prediction • Rating prediction • Success classification • Revenue forecasting • Sentiment analysis • Recommendation systems
Requirements: • Proper encoding • Feature scaling • Clean schema • No missing values

11. Deployment Requirements

The application should be production-ready. Deployment checklist: • requirements.txt • README.md • Streamlit deployment • GitHub repository • Modular code structure • Environment configuration • Performance optimization

12. Deliverables

Final deliverables must include: • Cleaned dataset • Jupyter notebook • Streamlit application • Source code • Dashboard screenshots • Business report • Requirements documentation • Deployment link • GitHub repository

13. Expected Outcomes

The project should provide: • Actionable business insights • Executive-level reporting • Professional analytics dashboard • Market intelligence capabilities • Interactive decision-making tools • Future ML integration support
The final solution should demonstrate: • Data analysis expertise • BI dashboard development • Data storytelling • Python engineering • Product analytics understanding
